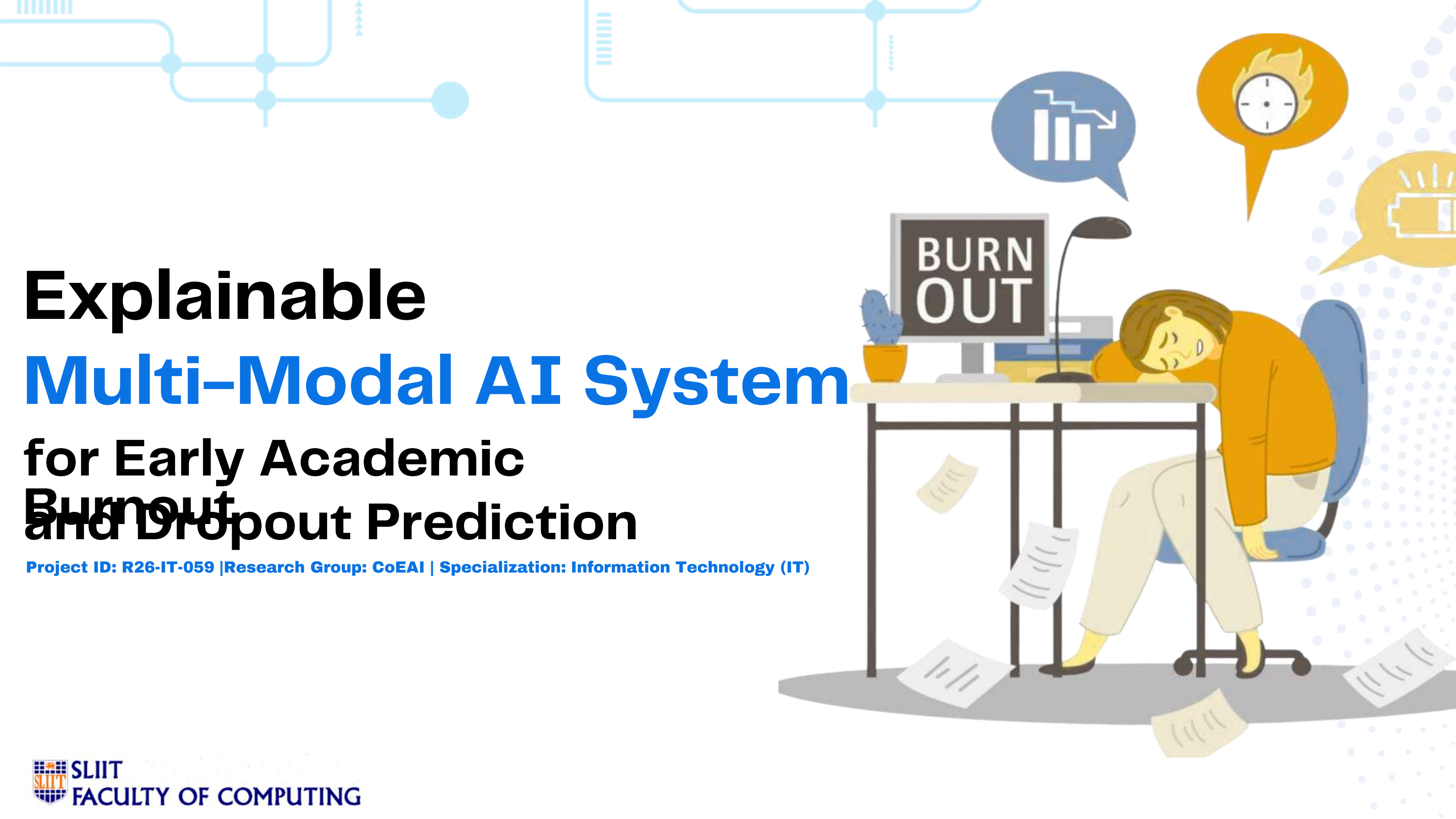
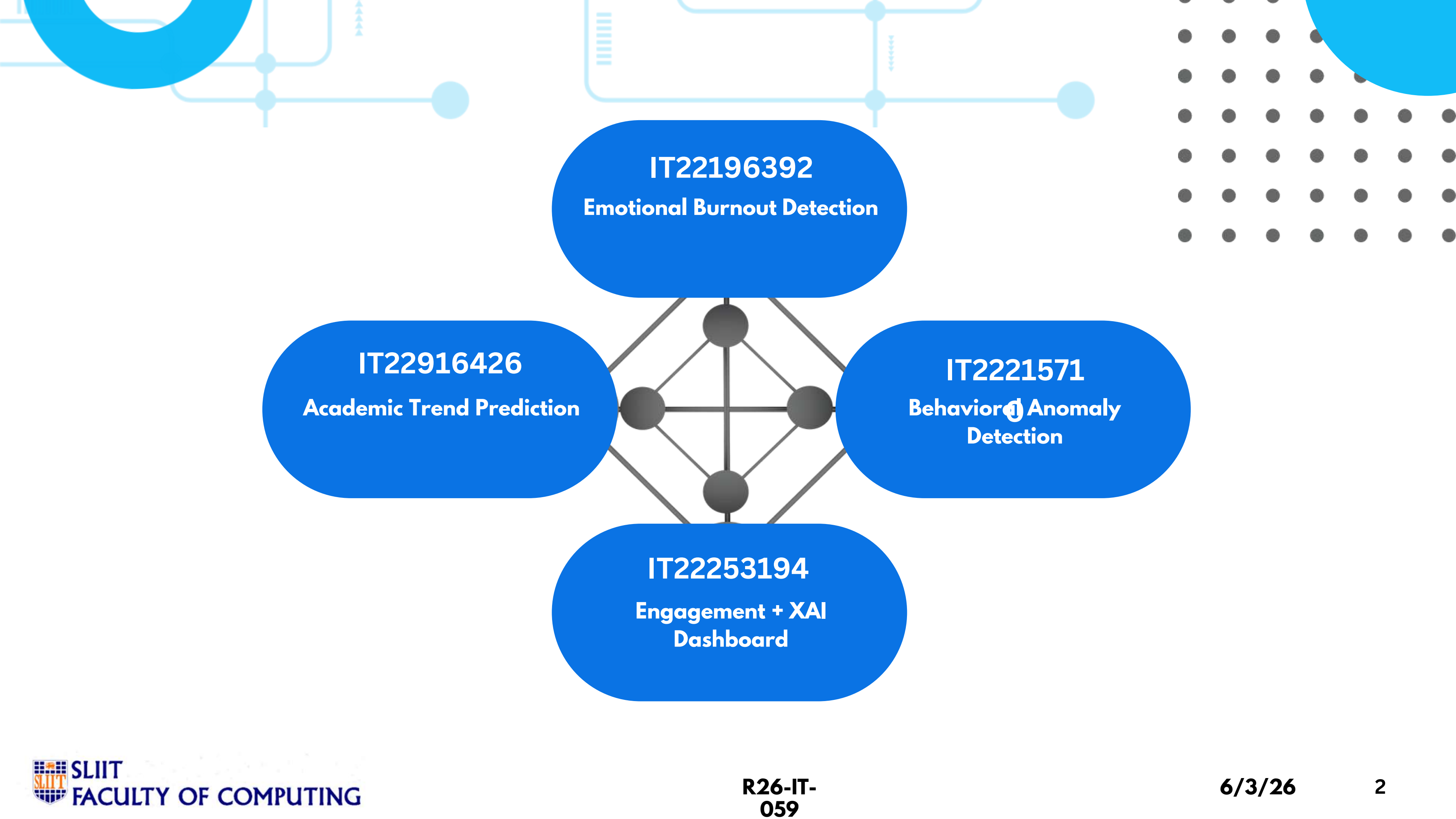




Explainable Multi-Modal AI System

for Early Academic Burnout and Dropout Prediction

Project ID: R26-IT-059 | Research Group: CoEAI | Specialization: Information Technology (IT)



IT22196392
Emotional Burnout Detection

IT22916426
Academic Trend Prediction

IT2221571
Behavioral Anomaly Detection

IT22253194
Engagement + XAI Dashboard

THE PROBLEM

Late Detection

Traditional systems detect burnout only after GPA drops, attendance fails, or formal complaints arise — too late for effective intervention.

No Continuous Monitoring

Universities lack passive, continuous monitoring. Rich digital signals in LMS, forums, and devices go unanalyzed.

Black-Box AI

Existing prediction systems lack explainability — limiting trust and adoption among educators and counselors.

Video Tutorial



USERS, STAKEHOLDERS & IMPACT

PRIMARY USERS

University Students

At-risk individuals experiencing burnout before it manifests as failure

Academic Advisors

Need early, interpretable alerts to intervene in time

Faculty Members

Monitor class-level engagement trends

SOCIETAL IMPACT

Reduced dropout rates

25-40% possible reduction with early intervention

Mental health support

Pre-emptive counseling before crisis stage

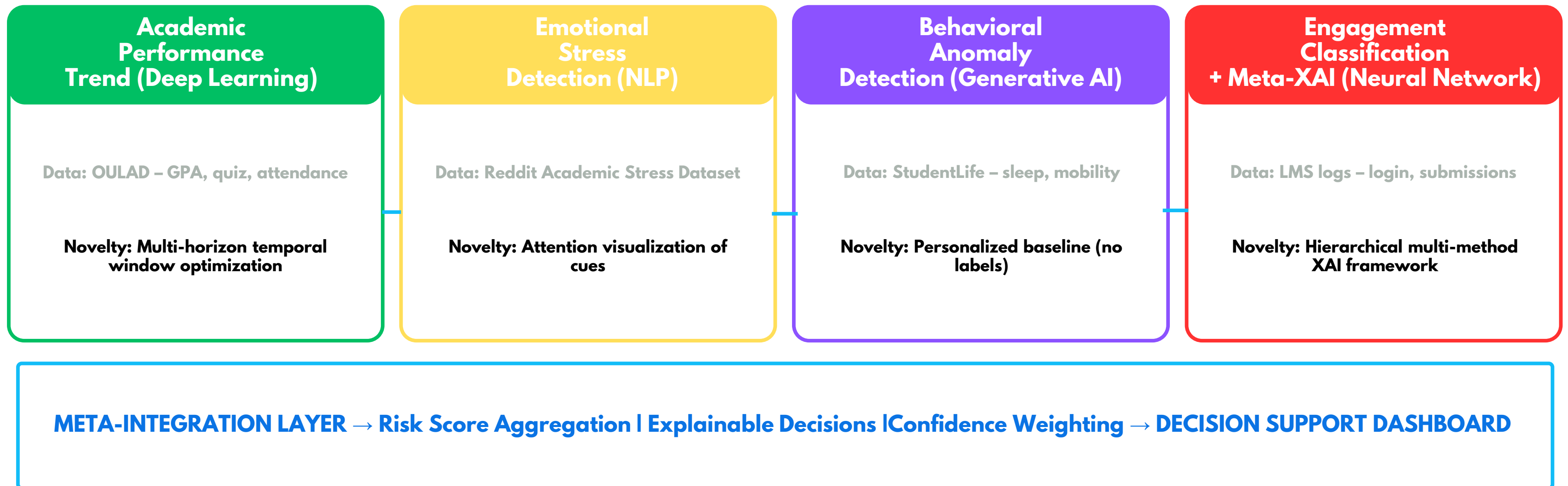
Institutional savings

Reduced cost of remedial programmes & re-enrollment

EXISTING SOLUTIONS & THEIR LIMITATIONS

APPROACH	TECHNOLOGY	ACCURACY	KEY GAP	REFERENCE
Academic Performance Prediction	LSTM on OULAD	75-85% acc.	Only academic data; no early warning	Waheed et al. (2023)
Text-Based Emotional Analysis	BERT fine-tuned	>90% acc.	No explainability; standalone model	Chen & Martinez (2023)
Behavioral Pattern Analysis	Mobile Sensing (StudentLife)	2-4 wk ahead	Small dataset (48 students); not scalable	Kumar et al. (2023)
Multi-Modal Integration	Academic + LMS logs	85-90% acc.	No XAI framework; no temporal opt.	Hellas et al. (2018)

PROPOSED SOLUTION OVERVIEW



Academic Performance Trend Prediction with Multi-Horizon Temporal Window Optimization

IT2291642

6

NOVELTY (35%)

- **Multi-timestep temporal prediction optimization** — identifies the **EARLIEST** reliable intervention window balancing accuracy with **actionability**.
- Unlike existing systems that predict current risk, this answers: 'How far ahead can we reliably predict academic decline?'
- **Feature-level explainability** reveals which signals (attendance vs. quiz scores vs. assignments) matter most at each prediction horizon.

RESEARCH OBJECTIVE

To develop a deep learning based academic performance trend prediction model with multi-horizon temporal window optimization and feature-level explainability to identify the optimal early intervention window for at-risk students using the OULAD dataset.

SUB-OBJECTIVES & TASKS

- **Collect & preprocess academic data from OULAD** (32,593 students, 22 courses)
- **Develop a deep learning time-series model** for academic trend prediction
- **Evaluate prediction accuracy** at multiple early warning timepoints (2–8 weeks ahead)
- **Integrate feature-level explainability** to identify key performance indicators at each prediction horizon
- **Contribute academic risk scores** to meta-integration layer



Academic Performance Trend Prediction with Multi-Horizon Temporal Window Optimization

IT2291642

6

APPLICATION OF DOMAIN KNOWLEDGE (30%)

Research Methodology

- Time-series deep learning for sequential academic pattern recognition
- Multi-horizon forecasting: predict risk at $t+2$, $t+4$, $t+6$, $t+8$ weeks
- Feature-level explainability for temporal prediction interpretation

Latest Technology

- PyTorch 2.0 recurrent deep learning with attention mechanism
- Additive feature attribution explainability for temporal models
 - Scikit-learn for baseline comparison (RF, XGBoost)

VALIDATION, DATA & ETHICS

Evaluation Metrics:	Precision, Recall, F1-Score, ROC-AUC at multiple lead times
Dataset	OULAD — CC-BY 4.0 licensed, 32,593 students, 22 courses, 10.6M interactions
Data Security:	Anonymized; no personally identifiable information used
Ethical Clearance:	Public dataset with open license; decision-support only (no automated penalties)

Emotional Burnout Detection from Student Text with Phrase-Level Interpretability

IT22196392

NOVELTY (35%)

- Attention mechanism visualization reveals WHICH specific words and phrases trigger burnout alerts.
- Enables advisors to understand the emotional context behind predictions — critical for trustworthy deployment.
- Predicts emotional exhaustion 4–6 weeks before academic decline becomes measurable (Sharma & Park, 2024).

RESEARCH OBJECTIVE

To develop a transfer learning based natural language processing model for detecting emotional burnout indicators from student-generated text, with phrase-level interpretability to identify specific emotional cues driving burnout risk.

SUB-OBJECTIVES & TASKS

- Collect & preprocess Reddit academic stress posts (r/college, r/CollegeRant, r/GetStudying)
- Fine-tune a transformer model for burnout-related multi-class emotion detection
 - Implement attention weight extraction for interpretability
 - Integrate text-based stress indicators into meta-learning
- Create emotional intensity scoring and trend visualization capabilities

Emotional Burnout Detection from Student Text with Phrase-Level Interpretability

IT22196392

APPLICATION OF DOMAIN KNOWLEDGE (30%)

Research Methodology

- Transfer learning: fine-tune a multilingual transformer model on domain-specific text
- Multi-class classification: frustration, exhaustion, disengagement, neutral
- Attention weight extraction for token-level interpretability

Latest Technology

- HuggingFace Transformers library with multilingual pre-trained model
- Custom attention visualization layer for phrase-level interpretability
- PyTorch 2.0 with GPU acceleration on Colab Pro

VALIDATION, DATA & ETHICS

Evaluation Metrics:	Precision, Recall, F1 (per emotion class), ROC-AUC; compare vs VADER, RoBERTa baseline
Dataset	Reddit Academic Stress Dataset — public posts ethically scraped via Reddit API / Kaggle
Data Security:	Text anonymized; usernames stripped; GDPR-compliant processing
Ethical Clearance:	Public forum data; no individual identification; decision-support only tool

Personalized Behavioral Anomaly Detection using Unsupervised Deep Learning with Individual Baseline Calibration

IT22215710

NOVELTY (35%)

- Personalized unsupervised burnout detection **WITHOUT** requiring sensitive mental health labels — preserving privacy.
- Establishes individual behavioral baselines; detects deviations specific to **EACH** student's normal patterns.
- Anomaly module explains **WHICH** behaviors changed: e.g., 'sleep dropped 40%, study sessions became irregular'.

RESEARCH OBJECTIVE

To develop an unsupervised deep generative model for behavioral anomaly detection that establishes personalized baselines from individual student patterns and detects burnout-associated deviations without requiring mental health labels.

SUB-OBJECTIVES & TASKS

- Preprocess StudentLife dataset (sleep, study duration, phone usage, mobility — 48 students, 10 weeks)
 - Develop an unsupervised deep generative model for behavioral pattern learning
- Implement personalized baseline learning — individual reconstruction error thresholds
- Create anomaly interpretation module identifying specific changed behaviors
- Integrate anomaly indicators with meta-integration layer



Personalized Behavioral Anomaly Detection using Unsupervised Deep Learning with Individual Baseline Calibration

IT22215710

APPLICATION OF DOMAIN KNOWLEDGE (30%)

Research Methodology

- Unsupervised deep generative modeling: reconstruction error as behavioral anomaly score
- Personalized thresholding: z-score baseline per student profile
- Feature attribution explainability on reconstruction components for behavioral dimension identification

Latest Technology

- PyTorch deep generative model with regularization for disentangled representation
- scikit-learn IsolationForest for baseline comparison
- Pandas + NumPy for time-series feature engineering

VALIDATION, DATA & ETHICS

Evaluation Metrics:	Anomaly detection F1, Precision@K, AUC; compare vs Isolation Forest & Autoencoder baseline
Dataset	StudentLife Dataset — Dartmouth College; academic research license; 48 students over 10 weeks
Data Security:	Mobile sensor data; no GPS stored; activity aggregated at hourly level; no direct identification
Ethical Clearance:	IRB-approved dataset; no mental health labels required; privacy-by-design approach

Engagement Classification & Hierarchical Explainable Meta-Integration System with Multi-Level Decision Transparency

IT22253194

NOVELTY (35%)

- Hierarchical multi-method explainability framework with **THREE** levels: (1) Model-specific — why each model flagged risk; (2) Meta-level — which modality contributed most; (3) User-centric — actionable advisor guidance.
- Directly addresses the 'black box' criticism preventing AI adoption in education — enabling trustworthy deployment.
- Neural network based meta-learner with confidence weighting enables dynamic modality prioritization per student profile.

RESEARCH OBJECTIVE

To develop a neural network based engagement classification model, hierarchical explainable meta-integration layer with multi-level decision transparency, and an interactive web dashboard translating multi-modal risk predictions into actionable advisor insights.

SUB-OBJECTIVES & TASKS

- Engineer features from LMS interaction logs (login frequency, content interaction, assignment submissions)
 - Develop neural network engagement classification model for dropout risk estimation
- Design meta-integration layer aggregating risk indicators from all four models
- Create unified XAI framework with 3-level interpretability (model, meta, user-centric)
- Design and implement an interactive web-based decision support dashboard

Engagement Classification & Hierarchical Explainable Meta-Integration System with Multi-Level Decision Transparency

IT22253194

APPLICATION OF DOMAIN KNOWLEDGE (30%)

Research Methodology

- **Ensemble meta-learning: stacking outputs from all four specialist models as meta-features**
- **Feature attribution explainability global and local analysis for meta-model transparency**
- **Three-tier XAI: model-level, meta-level, and user-facing dashboard**

Latest Technology

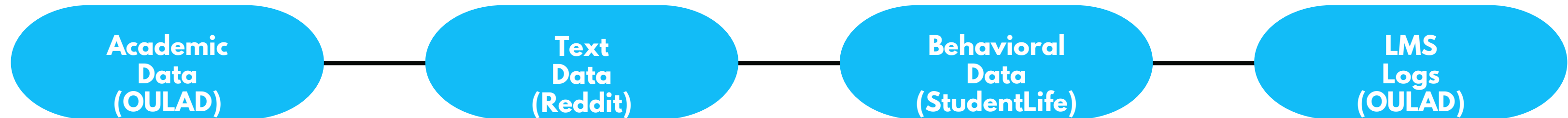
- **Flask + React for interactive web dashboard**
- **Model-agnostic explainability framework for black-box meta-model interpretation**
- **Plotly/Dash for temporal risk trajectory visualization**

VALIDATION, DATA & ETHICS

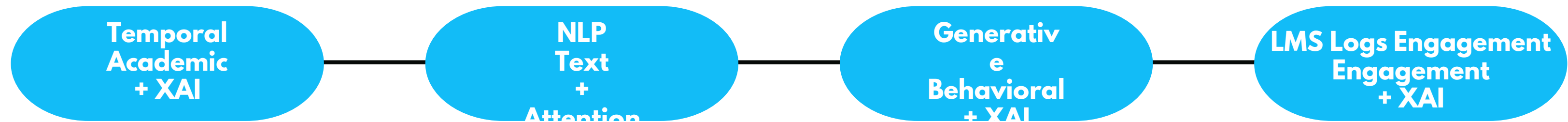
Evaluation Metrics:	System-level: F1, ROC-AUC, NDCG; XAI quality: explainability consistency, user study with advisors
Dataset	OULAD (LMS engagement) combined with outputs from Models 1–3; synthetic LMS data for stress testing
Data Security:	Aggregated risk scores only; no raw student data on dashboard; role-based access control
Ethical Clearance:	Decision-support only — human judgment required for ALL interventions; no automated actions

IMPLEMENTATION DETAILS — System Architecture

MULTI-MODAL DATA SOURCES



NEURAL MODEL ENSEMBLE LAYER



Explainable Multi-Modal AI for Early Academic Burnout & Dropout Prediction

Investment / Cost

- Labour: 4 students × LKR 75,000 = LKR 300,000
- Infrastructure: Google Colab Pro, AWS (approx. LKR 50,000)
 - Total: approx. LKR 350,000 (~USD 1,000)

Revenue / Pricing Model

- SaaS subscription: USD 5–15 per student/year
- Institutional license: USD 10,000–50,000/year per university
 - API licensing to EdTech platforms

IP Rights

- Research patent pending for temporal XAI optimization method
- Open-source dashboard under MIT license
- Commercial rights retained by SLIIT / team

Explainable Multi-Modal AI for Early Academic Burnout & Dropout Prediction

GANTT CHART & WORK BREAKDOWN
STRUCTURE

